Science of the Total Environment 723 (2020) 137995

Contents lists available at ScienceDirect

Science of the Total Environment

journal homepage: www.elsevier.com/locate/scitotenv

Investigation of hydrolysis acidification process during anaerobic treatment of coal gasification wastewater (CGW): Evolution of dissolved organic matter and biotoxicity

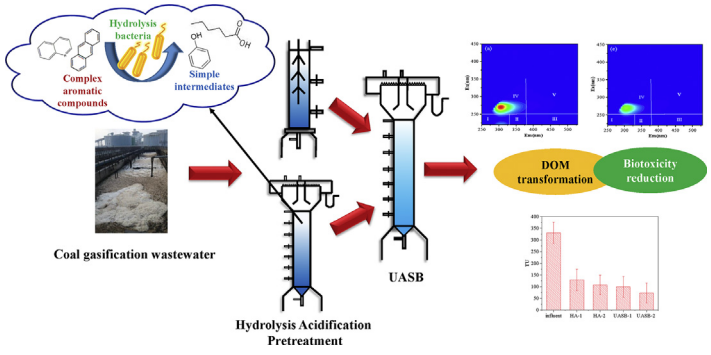
Zhiqiang Chen [a,b], Da Li [a], Qinxue Wen [a,*]

[a] State Key Laboratory of Urban Water Resource and Environment, School of Environment, Harbin Institute of Technology, Harbin 150090, China
[b] School of Civil Engineering, Lanzhou University of Technology, Lanzhou 730070, China

HIGHLIGHTS

- DOM transformation during bio-treatment for CGW was for the first time investigated.
- EEM-FRI and 2D-COS upon UV–Vis spectra were used for analysis.
- UASB reactor achieved a better hydrolysis efficiency compared to microaerobic hybrid reactor.
- Obvious "indirect estrogenic activity" after treatment should cause enough concern.
- Spectral indicators can be used to indicate DOM and acute toxicity removal.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:
Received 29 December 2019
Received in revised form 25 February 2020
Accepted 15 March 2020
Available online 17 March 2020

Editor: Paola Verlicchi

Keywords:
Coal gasification wastewater
Hydrolysis acidification
Dissolved organic matter
Biotoxicity
Pearson correlation analysis

ABSTRACT

Coal gasification wastewater (CGW) contains several types of aromatic pollutants, which impart high biotoxicity and reduce the quality of anaerobic treatment. Two types of hydrolysis acidification processes, namely microaerobic hybrid reactor (HA-1) and upflow anaerobic sludge blanket reactor (HA-2), were developed for pre-treatment before the anaerobic treatment. The changes in the dissolved organic matter and biotoxicity were investigated to comprehensively understand the degradation process. The results showed that HA-2 coupled with an anaerobic reactor achieved a 12.3% and 13.4% higher removal efficiency for chemical oxygen demand and total phenols, respectively, compared with the coupled process with HA-1. Furthermore, HA-2 could transform macromolecules into small molecules more efficiently and produce fewer intermediates. The coupled process with HA-2 preferentially removed complex aromatic substances with absorption wavelengths of 285 and 254 nm, according to the sequential orders interpreted from two-dimensional correlation spectroscopy. In addition, the results of fluorescence excitation-emission-matrix with regional integration analysis revealed that the contents of typical cyclic compounds in CGW, such as phenolic, heterocyclic, and polycyclic aromatic compounds were remarkably reduced by HA-2. In addition, HA-2 reduced the toxic unit value of CGW by 67.5% and increased the resazurin dehydrogenase activity of the sludge by 37.5% during CGW treatment, thus improving the biotoxicity removal and biodegradability. However, the coupled process with HA-2 did not significantly affect the "indirect estrogenic activity" of CGW. A Pearson correlation analysis indicated that spectral indicators, such as UV_{254} and $\Phi_{T,n}$, presented a high positive correlation with the reduction of acute toxicity and organics.

© 2020 Elsevier B.V. All rights reserved.

* Corresponding author at: School of Environment, Harbin Institute of Technology, Harbin 150090, China.
E-mail address: wqxshelly@hit.edu.cn (Q. Wen).

https://doi.org/10.1016/j.scitotenv.2020.137995
0048-9697/© 2020 Elsevier B.V. All rights reserved.

1. Introduction

Coal gasification is a clean and efficient energy-conversion method, and plays an important role in adjusting the energy structure, particularly regarding the situation in China of "rich coal, less oil, lack of gas" (Shi et al., 2019a). However, a large amount of wastewater is produced from the coal gasification process, containing high concentrations of refractory and toxic compounds, such as phenols, polycyclic aromatic hydrocarbons, nitrogen heterocyclic compounds, long chain n-alkanes, and thiocyanate, among others (Ma et al., 2018a; Zhao and Liu, 2016). The existence of hazardous compounds in coal gasification wastewater (CGW) constrains industrial development due to its adverse effect on environmental safety and public health.

In general, CGW is treated successively through the phenol and ammonia recovery process, biological treatment process, and advanced treatment process (Ji et al., 2016). Among these, anaerobic treatment is an indispensable process in which refractory compounds can be degraded or converted to easily biodegradable compounds, thus reducing the toxicity and improving the biodegradability of CGW (Fuchs et al., 2011; Meckenstock and Mouttaki, 2011; Shi et al., 2019b). However, for chemical oxygen demand (COD) and total phenols (TPh), a removal efficiency of only 20–30% has been achieved during the anaerobic treatment of CGW with conventional mesophilic upflow anaerobic sludge blanket (UASB) reactors (Wang et al., 2011). The hydrolysis acidification (HA) process has been widely used as a pre-treatment process for various industrial wastewaters (Song et al., 2019; Wang et al., 2014; Wu et al., 2016) to improve the wastewater-treatment process. In theory, HA can remove toxic compounds, change the composition of refractory organics, and reduce the concentration of pollutants in CGW, thus creating suitable conditions for the subsequent anaerobic process (Xu et al., 2015). However, the role of the HA process has not yet been sufficiently explained.

Moreover, UASB has been extensively used in the anaerobic treatment of various types of wastewater, particularly high-strength industrial wastewaters (Abbasi and Abbasi, 2012; Kong et al., 2019a); however, its long start-up period, susceptibility to shock loading, and high amount of suspended solids in the effluents, among others (Kong et al., 2019b), have motivated the innovation of many processes based on UASB. For example, anaerobic hybrid reactors, which combine the sludge bed in the lower zone and biofilm media in the upper zone, perform well in industrial wastewater treatment owing to the existence of both suspended granules and immobilized biofilm (Ramakrishnan and Surampalli, 2012; Show et al., 2020). In addition, micro-aeration treatment technology has been proved as an effective strategy to improve the anaerobic process in accelerating hydrolysis, increasing the inhibition of sulphate reduction, and reducing the massive accumulation of volatile fatty acids (VFAs), among others (Chen et al., 2020). Furthermore, microaerobic hydrolysis acidification has been successfully used as a pre-treatment process in petrochemical wastewater plants (Wu et al., 2016). Theoretically, the integration of micro-aeration and the anaerobic hybrid process developed in this research can improve the hydrolysis of CGW as the pre-treatment for the subsequent biological treatment.

The transformation of dissolved organic matter (DOM) during the wastewater treatment process not only reflects the removal efficiency of pollutants, but also reveals the reaction mechanism of wastewater treatment units to some extent, contributing to the selection and optimization of the combined process (Aftab and Hur, 2019; Chen et al., 2017a; Peng et al., 2018). Recently, spectroscopic methods, such as fluorescence excitation-emission-matrix (EEM) spectroscopy, ultraviolet-visible (UV–Vis) spectroscopy and Fourier transform infrared spectroscopy (FTIR), have been widely used to depict shifts in DOM. Furthermore, much more accurate and comprehensive information on the constituent change of DOM can be acquired with the assistance of powerful mathematical tools, including fluorescence regional integration (FRI) (Chen et al., 2003), parallel factor analysis (PARAFAC) (Stedmon and Bro, 2008), and two-dimensional correlation spectroscopy (2D-COS) (Phong and Hur, 2016). FRI and PARAFAC are EEM quantitative analysis methods, and 2D-COS method can be employed to reveal the direction and sequential order of DOM changes. For example, in a previous study, the degradation behaviour of effluent organic matter in ozonation, UV/H_2O_2, and UV/persulfate has been revealed by EEM-PARAFAC and 2D-COS coupled with gel permeation chromatography, and the reaction mechanism involved in the three advanced oxidation processes was comprehensively discussed (Chen et al., 2017b). Aftab et al. (2018) gained novel insights into the DOM oxidation behaviours of landfill leachate in Fenton oxidation processes under different operation conditions via EEM-PARAFAC and 2D-COS-FTIR. In summary, spectroscopic methods coupled with mathematical tools can be used to reveal the transformation of heterogeneous DOM in CGW, and can therefore explain the role of the HA process. In addition, some studies have indicated that there exists a correlation between the spectral indicators and biotoxicity, such as UV_{254} and the total fluorescence intensity (Tang et al., 2014; Zhang et al., 2019), suggesting that spectral indicators could be used to reveal the complex biotoxicity, hence, will demonstrate the efficiency of wastewater treatment, and the former can be obtained quickly and cost-effectively. Nevertheless, sufficient information regarding spectral indicators and biotoxicity in CGW is not available.

Although several technologies have been applied in CGW treatment to satisfy the national discharge standard, treated effluent has still been found to contain some recalcitrant and toxic pollutants, which may pose a potential risk to ecosystems and human health (Zhu et al., 2018). Various biotoxicity methods (acute toxicity, genotoxicity, estrogenic activity, and anti-estrogenic activity) (Ma et al., 2019a) should be used to obtain comprehensive information on the negative effects of the heterogeneous CGW. However, most studies on the biotoxicity of CGW have focused on the acute toxicity of CGW, and there is a lack of research for determining the toxic effects based on multi-methods (Ma et al., 2018b; Xu et al., 2019; Zheng et al., 2018). Because the composition of organic pollutants in CGW through various treatment processes is different, which may cause diverse toxic effects, exploring the evolution of biotoxicity in CGW anaerobic treatment could directly reveal the detoxification effects exerted by anaerobes, further providing the critical reference for the optimization of the anaerobic process and selection of the subsequent process.

In the present study, two types of HA methods coupled with UASB were developed for treating CGW. The main objectives of this study were to: (1) evaluate the efficiency of the CGW anaerobic process with different HA pre-treatments; (2) clarify the role of HA in enhancing the anaerobic CGW treatment by identifying the transformation of DOM; (3) explore the variations of the biotoxicity and biodegradability from influent to effluent.

2. Materials and methods

2.1. Characteristics of wastewater and inoculum

The wastewater used in this research was synthesized with crude phenol as the carbon source, which was obtained from the coal chemical industry in Yima, Henan Province. To improve the biodegradability of the wastewater, approximately 30% (v/v) domestic wastewater was added. NH_4Cl and KH_2PO_4 were used as the nitrogen and phosphorus source, respectively. To ensure the normal growth of microorganisms, mineral salts were added to satisfy the microbial needs for the micro and trace elements. The finial concentrations (mg/L) of the mineral salts in the synthetic wastewater were as follows: $CaCl_2 \cdot 2H_2O$ (20), $MgSO_4 \cdot 7H_2O$ (50), $FeSO_4 \cdot 7H_2O$ (15), $FeCl_3 \cdot 3H_2O$ (1), $MnCl_2 \cdot 4H_2O$ (0.5), $ZnCl_2$ (0.5), $CuCl_2$ (0.5), $(NH_4)_2MoO_4 \cdot 4H_2O$ (0.5), $AlCl_3$ (0.5), $CoCl_2 \cdot 2H_2O$ (0.5), $NaBO_2 \cdot 10H_2O$ (0.5), and $NiCl_2 \cdot 2H_2O$ (0.5). Moreover, $NaHCO_3$ was added to maintain the pH at 7–7.5. The key characteristics of the influent are listed in Table 1.

The inoculated anaerobic granular sludge was acquired from the anaerobic treatment unit of a wastewater treatment plant in Harbin (China). The total suspended solids (TSS) and volatile suspended solids (VSS) concentrations were 43 g TSS/L and 31 g VSS/L, respectively, and VSS/TSS was 0.72. One-third of the effective volumes of both the hydrolysis acidification reactors and the subsequent UASB reactors were inoculated with the sludge, and the final suspended sludge concentrations were all approximately 10 g VSS/L.

2.2. Experimental design

Two types of hydrolysis acidification reactors, namely a microaerobic hybrid reactor (HA-1, 600 mm in height, 90 mm in diameter, working volume of 3.5 L) and a UASB reactor (HA-2, 650 mm in height, 90 mm in diameter, working volume of 3.5 L), were coupled with two identical UASB reactors (UASB-1 and UASB-2, respectively; 1100 mm in height, 130 mm in diameter, working volume of 14 L) to treat the synthetic CGW (Fig. S1). The composite filler (150 mm in diameter) used in HA-1 comprises a plastic ring in the middle and some polyester fibre around it. Five such rings are fixed vertically in HA-1 at equal intervals at the upper part, occupying approximately 2/3 of the entire height of the reactor. To economically induce the microaerobic condition, the top of HA-1 was exposed to air, and the dissolved oxygen (DO) was maintained at 0.1–0.9 mg/L. The hydraulic retention times of the HAs and the subsequent UASBs were 12 h and 47 h, respectively. HA-1 and HA-2 were operated at room temperature, while UASB-1 and UASB-2 were operated at 35 ± 1 °C. All the reactors were fed from the bottom, and effluent recycling was conducted to maintain the upflow velocity at 0.5 m/h. The biogas of each UASB was collected using a gas sampling bag, and subsequently, the volume was measured using a 150-mL syringe.

To reduce the effect of the high toxicity of the influent, the two sets of reactors were initially fed with diluted synthetic CGW, and sucrose was used as an extra carbon source during the acclimation period. The COD concentration of the influent was increased step-by-step based on whether the effluent COD was steady. A comparative investigation of the performance between HA-1 and HA-2 was conducted at the steady state, including COD removal, TPh removal, and VFAs and pH variation. Additionally, the DOM transformation and biotoxicity reduction along the two sets of reactors were simultaneously determined. All the aforementioned parameters were measured in triplicate to ensure the validity and representativeness of the experimental results.

2.3. Analytical methods

The COD, NH_3-N, pH, TSS, and VSS were measured according to the standard methods (Chinese SEPAC, 2002). The TPh were determined by bromize-titration (Zhang et al., 2013). The composition of VFAs was analysed by gas chromatography (GC; GC 7890 N/FID, Agilent, USA) with flame ionization detection (Wen et al., 2018). The methane concentration in the biogas was also determined using GC (Agilent 6890A, USA) (Zhang et al., 2018). The composition of organics was analysed using GC-mass spectrometry (GC–MS) (6890GC-5973/5975MSD, Agilent, USA).

The dissolved organic carbon (DOC) of the samples was measured using a TOC-V_{CPH} analyser (SHIMADZU, Japan). The UV–Vis spectroscopy analysis was performed using a UV–VISIBLE Spectrophotometer (SHIMADZUUV-2550, Japan). The molecular weight (MW) distribution of the DOM was measured via high-performance liquid chromatography with a gel-filtration chromatography (GPC) column; the calibration curve of the GPC was obtained according to a range of polystyrene sulfonate standards (Li et al., 2019). Fluorescence EEM was measured using a fluorescence spectrophotometer (HITACHI F-7000, Japan). Prior to the EEM measurement, the samples were diluted to UV_{254} below 0.05 cm^{-1} to avoid the inner filter effect. Furthermore, some pre-treatment methods were used to remove the Raman scatter and Rayleigh scattering, to prepare for the FRI analysis (Chen et al., 2017a).

2.4. Biotoxicity and biodegradability

2.4.1. Acute toxicity test

The acute toxicity was assessed using a bioluminescence inhibition assay with a Microtox Model 500 analyser (Modern Water Inc., New Castle, USA). The marine bacterium *Vibrio fischeri* was used as the testing organism. The basic mode was employed to determine the effective concentration (EC_{50}) of each sample, which indicated that the bioluminescence of *Vibrio fischeri,* exposed to the samples, was reduced by 50% relative to a toxicant-free control. In brief, each sample was diluted to create four test concentrations, as instructed by the manufacturer, and Microtox Diluent was used as the toxicant-free control. First, the initial bioluminescence was measured, and then all the sample diluents were mixed with the bacterium, and finally, the bioluminescence was measured after 15 min to determine the EC_{50} for each sample. Furthermore, the toxic unit (TU) was calculated as the reciprocal of the EC_{50} to reflect the acute toxicity (Na et al., 2017).

2.4.2. Resazurin dehydrogenase activity (DHA)

Dehydrogenase activity is related to the activity of the microbial metabolism in activated sludge. Based on a previous report (Zhu et al., 2017), the DHA was measured using two types of activated sludge, one of which was taken from a full-scale municipal wastewater treatment plant, and the other from a bench-scale A/O-MBR for CGW treatment. Briefly, the activated sludge was mixed with samples and a phosphate buffer solution (PBS) (pH 7.4), and the PBS was regarded as the control. After 30 min, the DHA of the activated sludge was determined. In this study, the DHA of municipal sludge (M-S) could indicate the toxicity of the samples owing to its sensitivity towards CGW, while the DHA of sludge for CGW treatment (C—S), already adapted to toxic pollutants, could characterize the biodegradability of the samples.

2.4.3. Cytotoxicity and specific toxicity tests

The yeast cells after 4-h exposure to the samples were used to reveal the cytotoxicity of the samples. The detailed calculation steps can be found in a previous report (Chen et al., 2017a). Three types of specific toxicity were monitored, namely genotoxicity, estrogenic activity, and anti-estrogenic activity. Genotoxicity was analysed using SOS/umu-test with *Salmonella typhimurium TA1535/pSK1002* (Wang et al., 2007). As the positive control, 4-Nitroquinoline-Noxide (4-NQO) was applied to create the dose-response curve for characterizing the genotoxicity of the samples using the equivalent 4-NQO concentration. Estrogenic activity was measured using a yeast two-hybrid assay (Nishikawa

Table 1
Characteristics of effluent from different treatment units.

Operation units	COD (mg/L)	TPh (mg/L)	DOC (mg/L)	UV_{254} (cm^{-1})	SUVA ($L \cdot mg^{-1}\ m^{-1}$)	pH
influent	2004.2 ± 35.0	656.1 ± 20.2	619.00 ± 2.55	6.52 ± 0.27	1.05	7.38 ± 0.04
HA-1	1750.3 ± 44.6	551.6 ± 13.6	544.80 ± 4.53	5.42 ± 0.11	0.99	7.28 ± 0.03
HA-2	1567.1 ± 43.6	549.8 ± 14.4	521.20 ± 11.31	4.88 ± 0.10	0.94	7.25 ± 0.01
UASB-1	1505.7 ± 39.6	505.0 ± 12.6	433.80 ± 38.75	5.02 ± 0.28	1.16	7.32 ± 0.06
UASB-2	1258.6 ± 37.2	417.3 ± 12.7	417.00 ± 19.52	4.66 ± 0.16	1.12	7.42 ± 0.07

et al., 1999), and was expressed as the equivalent 17β-estradiol (E_2) concentration. The anti-estrogenic activity was measured using the yeast cells and determined by evaluating the inhibition of the samples to the β-galactosidase activity caused by E_2 (Wu et al., 2009). Finally, the anti-estrogenic activity of the samples was converted to the equivalent Tamoxifen and was used as the positive control.

2.5. Mathematical analysis

Using the Origin 2016 (OriginLab, USA), multi-peak Gaussian fitting of the GPC results was performed to calculate the peak areas for a quantitative analysis of the variation in the MW. 2D-COS spectra were obtained using the 2D Shige software (Kwansei-Gakuin University, Japan) based on the UV–Vis data (Phong and Hur, 2016), including a synchronous and an asynchronous map. The sequential order of the changes in the UV–Vis spectra could be obtained based on the two generated spectra; the interpretation procedure is described in detail elsewhere (Noda, 2012). EEM-FRI was used to calculate the integrated volume of the five fluorescence regions according to a previous method (Chen et al., 2003). The normalized regional volume and the corresponding percentage of region "i" could be obtained and are expressed as $\Phi_{i,n}$ (AU-nm^2) and $P_{i,n}$ (%), respectively; $\Phi_{T,n}$ is the sum of each normalized regional volume ($\Phi_{i,n}$). The Pearson correlation coefficient was calculated using the SPSS software (IBM, USA) to analyse the relationship between the spectral indicators and the biotoxicity.

3. Results and discussion

3.1. Performance of the two sets of HA-UASB

The basic properties of the influent and treated effluents during the steady operation phase are listed in Table 1. As shown in Table 1, HA-2 coupled with UASB-2 (HA-2-UASB) obtained 37.2% and 36.4% removal efficiencies of COD and TPh, respectively, which increased by 12.3% and 13.4%, respectively, compared to the coupled process with HA-1 (HA-1-UASB). The removal efficiencies of COD and TPh in HA-2 were 21.8% and 16.2%, while the corresponding removal efficiencies of HA-1 were 12.7% and 15.9%, respectively. HA-2 exhibited a better removal capacity of COD and TPh, resulting in higher removal of COD and TPh in the coupled process. The CGW contains many refractory and toxic compounds, inhibiting the activity of microbes. Generally, HA could decompose the complex pollutants and detoxify the wastewater (Song et al., 2019). The biodegradability of the wastewater was improved after HA, to ensure that the subsequent UASB would perform better. The TPh contents in the effluents of HA-1 (551.6 ± 13.6) and HA-2 (549.8 ± 14.4 mg/L) were very similar; however, more phenols were removed in UASB-2 than UASB-1. This might be attributed to the decomposition of some refractory pollutants, which produced phenols as the intermediates during anaerobic hydrolysis. In theory, HA-1 should perform better than HA-2 with the advantage of more biomass and micro-aeration stimulation. In contrast, HA-2 enhanced the removal of COD and TPh compared to HA-1, which might be due to two reasons. On the one hand, HA-2 obtained a better hydraulic condition, leading to a higher sludge bed and complete mixing of the sludge and wastewater. However, in HA-1, the suspended sludge could not mix well with the influent which was trapped at the bottom by the filler. On the other hand, more importantly, the long-term exposure of the CGW to oxygen in HA-1 would result in the generation of some refractory compounds (heterocyclic aromatic hydrocarbons, derivatives of benzene series, etc.) (Zhu et al., 2018). Although previous reports have indicated that micro-aeration could effectively promote the diversity and activity of hydrolytic and fermentative microorganisms in the AD process (Chen et al., 2020), the positive effects on the microbes by a small dose of oxygen seemed to be negligible for high-concentration CGW treatment when the negative effect of the water quality dominated. More attempts are required to explore the relation among the DO concentrations, microbial community structure, and CGW components to establish a robust conclusion of micro-aeration.

UV_{254} and the specific UV absorbance (SUVA) are commonly used to indicate the aromaticity of wastewater. The SUVA of the samples was determined by dividing UV_{254} with the DOC concentration. As shown in Table 1, the removal efficiencies of UV_{254} of HA-1 and HA-2 were 16.9% and 25.2%, respectively; the removal was further enhanced to 23.0% and 28.5%, respectively, by combining HA-1 and HA-2 with the UASB. Therefore, the UV_{254} removal during the coupled process depends on the performance of HA. After HA, the SUVA decreased and then increased in the UASB process. The transformation and degradation of complex aromatic matters with high aromaticity in the synthetic CGW caused the decline of the SUVA in the HA. However, only some small aromatic molecules with lower aromaticity were removed in the UASB, thus leading to the uptrend of the SUVA (Section 3.2.1).

The VFAs in the influent and effluent mainly comprised acetates, while other types of VFAs were negligible (data not shown). The concentration of acetates in the effluent of the two HA reactors was approximately 20 mg COD/L, which was lower than that in the influent. According to the previous reports, the VFAs concentration could increase if the HA process functioned well (Ma et al., 2019b; Wu et al., 2015). Thus, the activity of acidifying bacteria was inhibited in both the HA reactors. Only a slight difference was observed between the acetate concentrations in the effluents of the HA and UASB, indicating that the activity of methanogens utilizing acetates was also suppressed, which was in accordance with the methane production in the UASB. The methane production of the two UASBs fluctuated during the steady operation period and ranged between 10 and 30 mL/d. The pH of the influent and effluent was also investigated (Table 1); it remained approximately 7.3 in all the samples, irrespective of the VFA concentrations.

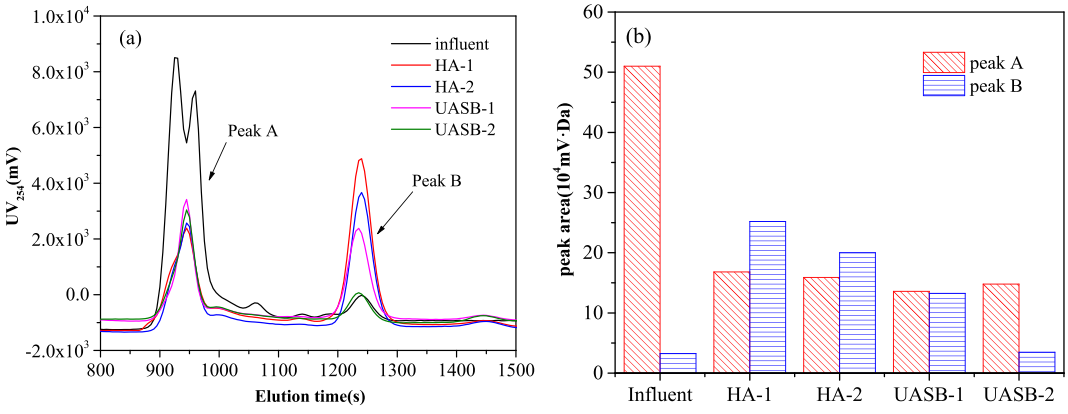


Fig. 1. (a) GPC results and (b) Gaussian-fitted peak areas of the samples from different units.

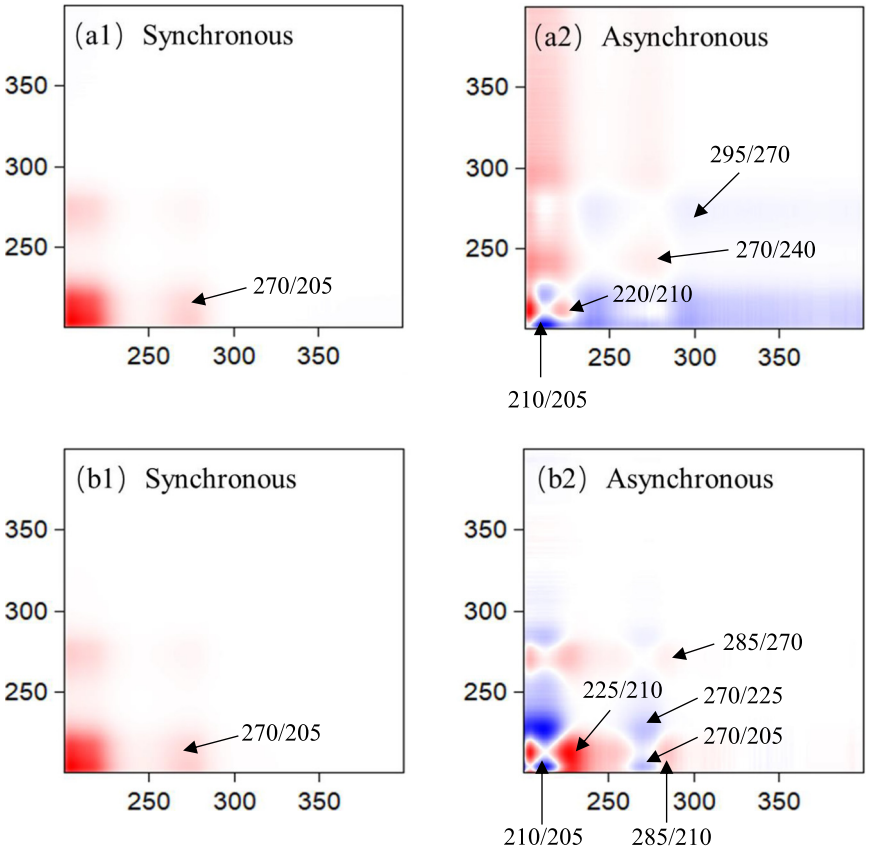


Fig. 2. Synchronous and asynchronous 2D-COS maps of UV–Vis spectra for the influent and treated effluent from (a) HA-1 and UASB-1, and (b) HA-2 and UASB-2.

The phenols, ammonia nitrogen, acetates, and sodium bicarbonate contained in the influent formed a buffer to avoid the fluctuation of the pH.

3.2. Transformation of the dissolved organic matter

3.2.1. Change of molecular weight

GPC was used to characterize the MW change of the DOM in the coupled process. As shown in Fig. 1a, two main peaks were determined with elution times of 940 s and 1235 s, representing 7654 Da (peak A) and 107 Da (peak B), respectively. Generally, peak A represents high-MW organics, like high-MW humic substances; however, the synthetic CGW used in this research mainly comprised aromatic pollutants, such as phenols and polycyclic aromatic hydrocarbons, but not humic substances. Considering the intermolecular forces between aromatic compounds (e.g. hydrogen bonding, π-π interactions, etc.), particularly for phenols (Vojta and Vazdar, 2014), it was assumed that peak A in this research represented a mixture of aromatic compounds, connected via multiple intermolecular forces, which showed the characteristics of large molecular weight and high aromaticity. Peak B corresponds with low-MW organics, such as aldehydes, alcohols, and phenols.

To quantify the changes in the MW, multi-peak Gaussian fitting was used to integrate the peak area using Origin 2016 (Fig. S2). The variation of the peak area of the samples is shown in Fig. 1b. The major UV_{254} of the influent was sourced from the aromatic mixture (peak A), as shown in Fig. 1b. After the HA process, the area of peak A decreased sharply from 51.0×10^4 mV Da to 16.8×10^4 mV Da and 15.9×10^4 mV Da in HA-1 and HA-2, respectively, and then changed slightly during the UASB process. Meanwhile, the area of peak B initially increased from 3.2×10^4 mV Da to 25.2×10^4 mV Da and 20.0×10^4 mV Da in HA-1 and HA-2, respectively, and then decreased to 13.3×10^4 mV Da and 3.5×10^4 mV Da in UASB-1 and UASB-2. It was evident that the high-MW molecules (peak A) were metabolized and degraded by microorganisms into relatively low-MW molecules (peak B) during the HA process. The present result is consistent with the literature that HA can efficiently transform most high-MW DOM (>100 kDa) in real domestic wastewater into low-MW DOM (<500 Da) (Liu et al., 2016). Theoretically, for a higher proportion of low-MW matter, the biodegradability of the wastewater would improve. The good biodegradability of wastewater was closely related to the fact that small molecules could gain access into bacterial cells more easily than in high-MW matter (Liu et al., 2016). Therefore, the effluent of the HA-1 process containing more low-MW matter should be more suitable for the subsequent anaerobic treatment. However, more small molecules (peak B) were reduced in UASB-2, although less low-MW matter (peak B) was present in the effluent of HA-2. A possible reason for this is that the low-MW matter in HA-1 may predominantly comprise intermediates, such as alcohols, ethers, and ketones, while the accumulation of metabolites tended to inhibit the anaerobic digestion process (Chen et al., 2008). Therefore, HA-2 could more efficiently transform macromolecules into small molecules and enhanced the biodegradability of wastewater, resulting in the overall improvement of the anaerobic treatment for CGW.

3.2.2. Change of UV–Vis spectroscopy

The variations of the UV–Vis spectra of the DOM in the influent and effluent of the coupled process are shown in Fig. S3. The absorbance intensities consistently decreased along the coupled process, centred at a specific wavelength range of 200–300 nm. Based on the spectral shapes, two absorbance bands could be distinguished in the wavelength ranges of 200–250 nm and 250–300 nm. To further enhance the spectral resolution, 2D-COS based on the UV–Vis spectra of the samples from different processes was applied. It was anticipated that the 2D-COS maps would identify the sequential changes of the UV–Vis spectra from

different anaerobic processes, and further clarify the difference in the removal behaviour of the UV-absorbing compounds between the two coupled processes. The synchronous and asynchronous maps generated by 2D-COS are shown in Fig. 2.

Two conspicuous auto-peaks were identified along the diagonal of both synchronous maps at wavelengths of 205 nm and 270 nm, and a positive cross-peak was found at 270/205 nm (Figs. 2 a1 and b1). The positive signs of the peaks denote the same direction of the spectral changes in the UV-absorbing compounds related to the peaks (i.e., absorbance decreased consistently along with the coupled process in this study). The intensity of the auto-peak shows the overall susceptibility of the spectra with respect to the different wavelengths (Guo et al., 2019). The intensities of the auto-peaks located at 205, 270, and 254 nm (not apparent) of the two synchronous maps were compared and followed the order: 205 nm > 270 nm > 254 nm, suggesting that the degradation of the DOM occurred predominantly at the wavelength of 205 nm. The asynchronous map further revealed the order of the pollutant removal at different wavelengths. For HA-1-UASB, two negatively signed cross-peaks (210/205 nm, 295/270 nm) and two positively signed cross-peaks (220/210 nm, 270/240 nm) were observed (Fig. 2 a2). Based on Noda's rule and the relevant peak signs (Table S1), the absorbance changed, following the order 205 nm > 220 nm > 210 nm > 270 nm > 254 nm > 240 nm > 295 nm. However, for HA-2-UASB, three negative cross-peaks (210/205 nm, 270/225 nm, 270/205 nm) and three positive cross-peaks (225/210 nm, 285/270 nm, 285/210 nm) were observed in the asynchronous map (Fig. 2 b2). According to Noda's rule and the relevant peak signs (Table S1), the order of the preferential reaction wavelength followed the order: 285 nm > 254 nm > 225 nm > 205 nm > 210 nm > 270 nm. Based on the previous reports, the chromophores containing π bonds (e.g. C═C and C═O) combined with benzene could induce strong UV absorption at 200–250 nm, and the absorbance at 250–280 nm was associated with aromatic structures like phenolic compounds (Guo et al., 2019; Hur et al., 2011). In addition, a larger π bond could be produced by the interaction between π–π or the π–π* conjugative effect, causing a redshift of λ_{max} (Guo et al., 2019). Therefore, the absorption wavelength at 280–300 nm was likely related to the complex aromatic compounds, such as polycyclic aromatic hydrocarbons and quinones. According to Section 3.2.1, UV_{254} mainly comprised aromatic macromolecules in the influent. By comparing the results between the two coupled processes, it was found that most of the DOM-containing benzene rings (205 nm) were reduced during the coupled process. HA-2-UASB had the advantage of removing complex aromatic compounds associated with the wavelengths of 285 and 254 nm, compared with HA-1-UASB.

3.2.3. Change of fluorescence

EEM fluorescence spectra are highly sensitive and selective, as well as non-destructive for samples. Hence, they have been widely used to characterize the DOM in both wastewater and natural water systems (Korak et al., 2014). The EEM fluorescence spectra of all the samples are presented in Fig. 3. Two evident fluorescence peaks were identified from the fluorescence spectra of the influent. The first peak (peak 1) was observed at excitation/emission wavelengths (Ex/Em) of 270/305 nm; it is related to the π–π conjugated groups of aromatic substances (e.g. phenols and benzoquinone) (Peng et al., 2018). The second peak (peak 2) appeared at Ex/Em of 220/300–310 nm, and is associated with aromatic matter-containing benzene rings (Peng et al., 2018). The dominant fluorescent peaks indicate that the phenols, benzene, and their derivates were the primary pollutants in the influent, which is in line with previous reports (Zhu et al., 2018).

To quantitatively assess the changes of the EEM spectra and fully apply the fluorescence data, the FRI technique was used to analyse the variation of the five fluorescence regions (Table 2). Soluble microbial by-products (region IV) and tyrosine-like and tryptophan-like aromatic proteins (regions I and II) were found to be the main chromophores, and the sum of the regional volume ($\Phi_{i,n}$) accounted for >90% of the DOM. As mentioned above, regions IV and I contained peaks 1 and 2; thus, are associated with phenols, benzene, and their derivates. Region II represents some nitrogen heterocyclic compounds (e.g. indole) (Carstea et al., 2016). In addition, regions III and V, commonly associated with fulvic acid-like and humic acid-like substances, represent to some

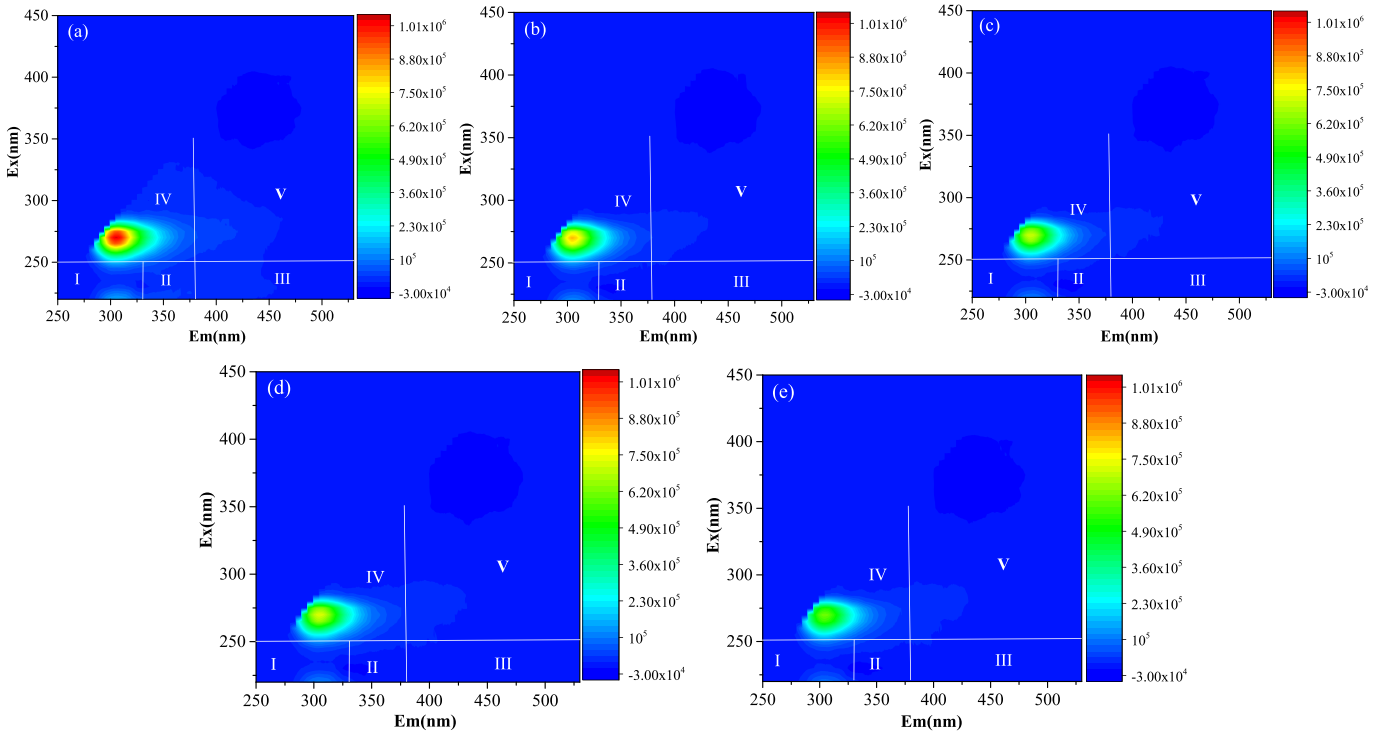


Fig. 3. The EEM fluorescence spectra and FRI distribution of the influent and treated effluent: (a) influent; (b) HA-1; (c) HA-2; (d) UASB-1; (e) UASB-2. Regions I and II: tyrosine-like and tryptophan-like aromatic protein; Region III: fulvic acid-like substances; Region IV: soluble microbial by-products; Region V: humic acid-like substances.

extent the polycyclic aromatic hydrocarbons, such as phenanthrene and fluoranthene (Carstea et al., 2016). Therefore, the typical cyclic compounds (i.e. phenols, nitrogen heterocyclic compounds, and polycyclic aromatic hydrocarbons) in the CGW could be analysed more comprehensively using FRI than the method involving searching for the peaks located at specific positions. Furthermore, region IV increased from 64.78% to >70% after the HA process, implying that intermediates were generated during the degradation of aromatic compounds (e.g. organic acids, ketones, and quinones) (Peng et al., 2018). As shown in Table 2, the sums of each integrated regional volume ($\Phi_{T,n}$) were reduced by 37.6% and 47.8% in HA-1 and HA-2, respectively. Then, after the UASB process, the removal of $\Phi_{T,n}$ increased to 44.1% and 52.5%, respectively, indicating that the HA process is essential to achieve the effective removal of typical aromatic compounds in CGW. Additionally, the $\Phi_{i,n}$ values of the five regions were all reduced by some degree during the HA process; whereas, only region IV was clearly decreased after the UASB process, indicating that HA played a key role in the transformation and removal of refractory aromatic pollutants during the coupled process, such as nitrogen heterocyclic compounds and polycyclic aromatic hydrocarbons. Overall, compared to HA-1, HA-2 removed more typical cyclic compounds in the synthetic CGW in the coupled process, particularly for refractory aromatic pollutants.

3.2.4. Identification of organic compounds

To confirm the speculation on the transformation and removal of the DOM in the synthetic CGW along the coupled process, the organic compounds in the influent and treated effluent were analysed by GC–MS. As shown in Table 3, the major organic compounds in the influent and treated effluent were phenols and their derivatives, accounting for >50% of the total peak area, while there were few heterocyclic compounds, as well as polycyclic aromatic hydrocarbons, in each sample. A possible reason for this is that there were only small amounts of heterocyclic compounds and polycyclic aromatic hydrocarbons in the water sample, or that the extraction force of the extractant to those contaminants was limited. Furthermore, some kinds of phenols were found to be totally removed in the coupled process; however, some new contaminants were present in the treated effluent (i.e. *p*-cresol and phenol, 2-ethyl), indicating the transformation of the DOM in the coupled process. Such a transformation most probably occurred in the HA process, based on the analysis of the MW, and UV–Vis and EEM spectra. Simultaneously, some ester and long-chain hydrocarbons were generated as intermediates in the treated effluent, and more intermediates were detected in the effluent of UASB-1 compared to that of UASB-2, which was coincident with the results of the MW distribution analysis, showing that the contaminants were more thoroughly degraded in the coupled process with HA-2. The spectra and chromatography combined with advanced mathematic tools could provide further insights into the DOM transformation in the CGW in the coupled processes.

3.3. Assessment of biotoxicity and biodegradability

3.3.1. Acute toxicity

The results of the acute toxicity of all the water samples of the coupled processes are depicted in Fig. 4. The TU of the influent to *Vibrio fischeri* decreased from 330 to 129 and 107 when subjected to the HA-1 and HA-2 process, respectively, and then reduced to 99 and 73 after the UASB process. HA-2-UASB achieved 77.9% removal of TU, which was higher than that by HA-1-UASB (70.0%). The highest reduction in TU occurred in the HA process, where the removal efficiency was 67.5%. Moreover, the high biotoxicity mainly resulted from the polar organic compounds in the CGW, including phenolic, heterocyclic, and polycyclic aromatic compounds, based on the previous literature (Ma et al., 2018b). Interestingly, although the phenols as the main pollutants were reduced more in the UASB-2 during the coupled process with HA-2, HA-2 still acted as the main contributor to the TU reduction, which reconfirmed to an extent that HA was able to achieve the degradation and transformation of refractory aromatic compounds besides phenols. Simultaneously, the removal of various aromatic compounds in the HA process might destroy the synergistic toxicity that existed among the multiple organic pollutants (Novak et al., 2017). Furthermore, based on the toxicity classification (Persoone et al., 2003), the

Table 3
The organic compounds compositions of influent and treated effluent.

Organic compounds	influent	UASB-1	UASB-2
Phenol	8.28[a]	2.85	4.76
Phenol, 2-ethyl	ND[b]	6.70	10.64
Phenol, 2-methyl	11.50	ND	10.34
Phenol, 2,4-dimethyl	3.81	10.78	4.35
p-Cresol	ND	11.57	16.00
Phenol, 4-ethyl	14.83	15.42	14.25
Phenol, 3,4-dimethyl	11.63	ND	ND
Phenol, 2,3,6-trimethyl	0.65	0.25	ND
Phenol, 4-propyl	0.71	ND	ND
Phenol, 2-(1-methylethyl)-	0.97	0.73	1.01
p-Cumenol	1.16	ND	ND
Phenol, 3-ethyl-5-methyl	0.95	1.76	2.47
Phenol, 3,4,5-trimethyl	0.77	1.06	1.01
Phenol, 2,4,6-trimethyl	1.17	0.73	1.91
Phenol, 2-ethyl-4-methyl	0.42	0.40	0.60
Benzene, pentamethyl	0.30	0.32	0.23
2,5-Cyclohexadiene-1,4-dione,2,6-dimethyl	0.21	0.12	0.19
Ethanone, 1-(4-ethylphenyl)-	ND	ND	0.66
Furan, 3-phenyl	ND	ND	0.36
Benzene, 1-ethyl-4-methoxy	3.96	3.57	3.79
1H-Inden-1-ol, 2,3-dihydro	1.24	1.28	1.89
1H-Inden-5-ol, 2,3-dihydro	1.14	1.14	1.56
6-Methyl-4-indanol	0.19	1.95	1.59
Dibutyl phthalate	ND	0.31	ND
Hexanedioic acid,dioctyl ester	0.21	ND	0.37
1,4-Benzenedicarboxylicacid,bis(2-ethylhexyl) ester	0.22	ND	0.24
1-Tetracosene	ND	5.10	ND
1-Hexacosene	ND	1.15	ND
Pentanedioic acid,(2,4-di-t-butylphenyl) monoester	ND	0.22	0.32
o-Hydroxybiphenyl	ND	0.18	ND
1-Naphthalenol, 2-methyl	ND	0.16	0.24
Phthalic acid, isobutyl4-octyl ester	ND	0.27	ND
Nonadecane	ND	0.21	ND
Tetracosane	ND	4.03	ND
Octane, 5-ethyl-2-methyl	ND	1.43	ND
others	35.68	26.31	21.22

[a] Values represent the relative percentage of total peak area.
[b] ND, not detect.

Table 2
Fluorescence regional integration for different EEM regions.

EEM region	Influent		HA-1		HA-2		UASB-1		UASB-2	
	$P_{i,n}$ (%)	$\Phi_{i,n}$($\times10^6$AU-nm^2)	$P_{i,n}$ (%)	$\Phi_{i,n}$($\times10^6$AU-nm^2)	$P_{i,n}$ (%)	$\Phi_{i,n}$($\times10^6$AU-nm^2)	$P_{i,n}$ (%)	$\Phi_{i,n}$($\times10^6$AU-nm^2)	$P_{i,n}$ (%)	$\Phi_{i,n}$($\times10^6$AU-nm^2)
I	15.66	1647.56	18.22	1197.43	17.84	979.39	17.69	1040.39	17.91	895.52
II	13.32	1402.11	6.17	405.47	5.44	298.98	6.33	372.31	6.23	311.49
III	4.94	519.86	2.64	173.24	2.33	127.88	2.66	156.3	3.62	181.33
IV	64.78	6816.75	73.15	4806.98	74.93	4114.45	73.56	4326.96	72.4	3620.89
V	1.3	136.77	−0.18	−12.19	−0.54	−29.63	−0.24	−13.84	−0.16	−7.91
Total	100	10,523.05	100	6570.93	100	5491.07	100	5882.12	100	5001.32

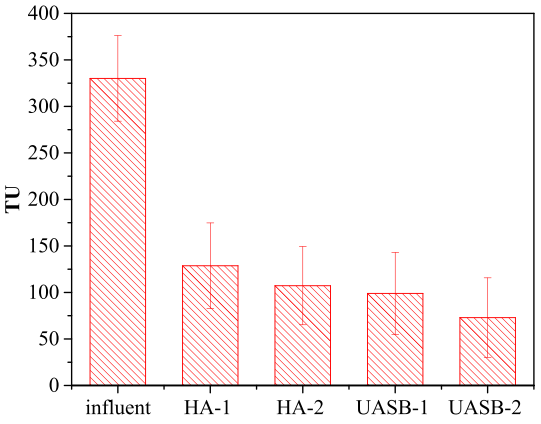


Fig. 4. Acute toxicity results of influent and treated effluent.

influent and the effluent of the HA were classified to have very high acute toxicity and the effluent of UASB had high acute toxicity, indicating that further treatment was necessary to mitigate the potential risks to the natural environment.

3.3.2. Evolution of DHA

Fig. 5 shows the variation of the DHA of all the samples using two types of activated sludge. The initial DHA of C—S was lower than that of M-S due to the long-term inhibition of toxicants in the CGW. When exposed to the influent, a sharp decrease was observed in the DHA of M-S, while the DHA of C—S increased because the C—S may have used parts of the toxic and refractory compounds as a carbon source. Similar to the acute toxicity evolution, the DHA of C—S and M-S clearly increased after the HA process, which could be attributed to the effective reduction of typical cyclic compounds in the influent, including phenols, and heterocyclic and polycyclic aromatic compounds and their derivates (Fig. 3). However, it should be noted that the DHA of the C—S and M-S slightly decreased in the UASB process, which may have been due to the degradation of biodegradable phenols, according to a previous report (Zheng et al., 2019). Compared with the DHA in the influent, the maximum improvement was observed after the HA-2 process; the corresponding increments were 37.5% (C—S) and 2020% (M-S). Therefore, HA played a leading role in reducing the toxicity of the CGW and improving its biodegradability.

3.3.3. Cytotoxicity and specific toxicity

To further investigate the potential risks to the environment and human health, the cytotoxicity and three kinds of specific toxicity

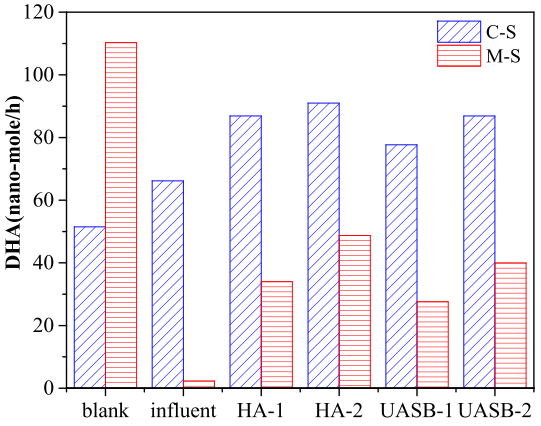


Fig. 5. Evolution of dehydrogenase activity for influent and treated effluent with two types of activated sludge. C-S: Activated sludge used for the CGW treatment; M-S: Activated sludge used for municipal wastewater treatment.

Table 4
Evolution of biotoxicity parameters (cytotoxicity, genotoxicity, estrogenic activity and anti-estrogenic activity) for influent and treated effluent.

Operation Units	Cytotoxicity	Genotoxicity	Estrogenic activity	Anti-estrogenic activity
	Effect (I_{yeast})	4-NQO (μg/L)	E2 (μg/L)	TAM (mg/L)
Influent	1.05	ND	ND	ND
HA-1	0.99	ND	ND	ND
HA-2	0.96	ND	ND	ND
UASB-1	0.94	ND	ND	ND
UASB-2	0.91	ND	ND	ND

Note: The units of genotoxicity, estrogenic activity and anti-estrogenic activity were used as equivalent concentration of reference substance.
The genotoxicity, estrogenic activity and anti-estrogenic activity have been tested for gradient dilution and concentration samples to guarantee the validity, and the maximum dilution and concentration ratios are 10000 and 100, respectively.

were used to evaluate the biotoxicity of the effluent from different processes. The bioluminescence inhibition assay and cytotoxicity both belong to nonspecific toxicity, and the latter is usually used to estimate the toxicity of disinfection byproducts formed during chlorination (Du et al., 2018). According to the cytotoxicity tests, there was no obvious inhibition of the samples on yeast cells (Table 4), though the sample exhibited acute toxicity to luminous bacteria under the same condition. It was suggested that luminous bacteria were more sensitive than yeast cells on the detection of the nonspecific toxicity for the CGW. Simultaneously, it was clear that the genotoxicity, estrogenic activity and anti-estrogenic activity were not all detected (Table 4). However, it has been reported that the raw CGW showed evident genotoxicity by damaging the DNA, which was proved by the comet assay using *Tetrahymena thermophila* as the testing organism (Ma et al., 2018b). In this study, the genotoxicity was detected via the SOS/umu-test, which was based on a series of biochemical reactions, including the expression of a specific gene. It was possible that the specific reaction was inhibited or destroyed by the toxicants in the CGW, resulting in the undetectable genotoxicity. As for the estrogenic activity or anti-estrogenic activity, synergistic or antagonistic effects induced by the various pollutants in the CGW might explain the results. Interestingly, the estrogenic activity was detected when the samples were mixed with E_2 (Fig. S4). As

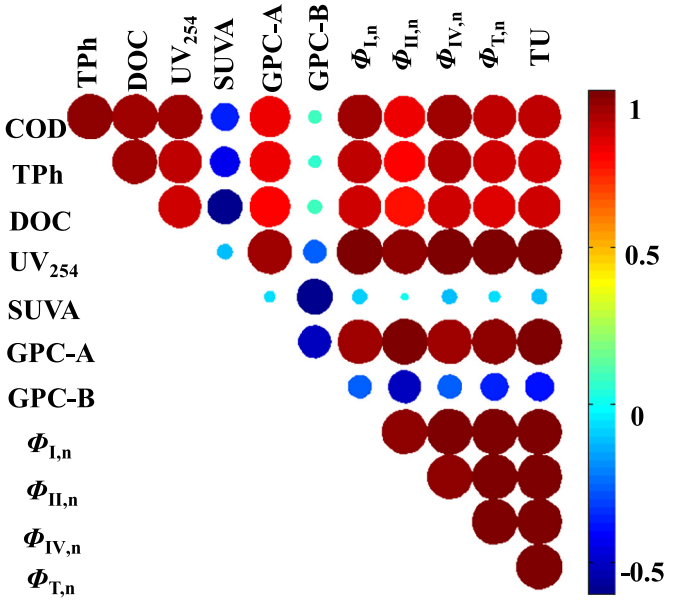


Fig. 6. Correlation matrix between the regular parameters, spectral indicators, and acute toxicity. GPC-A: Chromatogram area of fitted peak A. GPC-B: Chromatogram area of fitted peak B.

known, I_g was used to determine the antiestrogenic activity, which was generally positive (Wu et al., 2009). However, the I_g value of each sample was negative (Fig. S4), indicating that the samples could promote the estrogenic activity induced by E_2; this activity was named "indirect estrogenic activity" in this study. Furthermore, apparent "indirect estrogenic activity" still existed in the effluent of the coupled process, which should be continually investigated during the subsequent processes to ensure environmental safety. More efforts are required to explore the specific toxicity of CGW to obtain a robust conclusion, such as trying more extensive dilution or concentration range of the samples to determine the toxicity.

3.4. Pairwise correlations between regular parameters, spectral indicators and acute toxicity

The relationships between the regular parameters, spectral indicators, and acute toxicity were determined by the Pearson correlation matrix and are depicted in Fig. 6. The detailed data are listed in Table S2. The results showed that the regular parameters (i.e. COD, TPh, and DOC) were positively correlated with each other ($R = 0.946–0.974$, $P < .05$), indicating that the phenolic compounds were the major pollutants in the CGW. Meanwhile, the COD was positively associated with UV_{254}, $\Phi_{I,n}$, $\Phi_{IV,n}$, $\Phi_{T,n}$, and TU ($R = 0.882–0.974$, $P < .05$), and TPh was positively related to these parameters, except $\Phi_{T,n}$ and TU ($P > .05$), indicating that the CGW comprised various aromatic compounds besides phenols, and that the reduction of the acute toxicity was attributed to the removal of all the organics in the CGW, not limited to the phenols removal. Furthermore, it was found that UV_{254}, the chromatogram area of fitted peak A (GPC-A), $\Phi_{I,n}$, $\Phi_{II,n}$, $\Phi_{IV,n}$, and $\Phi_{T,n}$ were positively correlated ($R = 0.939–1.000$, $P < .05$), suggesting the unsaturated matters in CGW were related to aromatic pollutants, and the hypothesis that GPC-A was the assemblage of various aromatic substances based on multiple intermolecular forces could be proved to some degree. In addition, it is worth noting that the TU had a strong and positive correlation with UV_{254}, GPC-A, $\Phi_{I,n}$, $\Phi_{II,n}$, $\Phi_{IV,n}$, and $\Phi_{T,n}$ ($R = 0.976–0.993$, $P < .01$), and the highest relationship appeared between $\Phi_{T,n}$ and TU, implying that the acute toxicity was mainly caused by the aromatic matters in the CGW. Therefore, it was clear that the variation of the regular parameters and acute toxicity during the anaerobic treatment of the CGW could be adequately depicted by the spectral indicators, particularly UV_{254} and $\Phi_{T,n}$.

4. Conclusion

HA played a key role in the DOM transformation, wastewater detoxification, and biodegradability improvement. Compared to HA-1, HA-2-UASB achieved a better performance regarding the removal of organic pollutants. Moreover, HA-2 had an advantage over HA-1 in transforming the DOM, which was mainly reflected in three aspects: (1) HA-2 converted more macromolecules into small molecules with less intermediates; (2) with the help of HA-2, the coupled process preferentially removed complex aromatic substances with absorption wavelengths of 285 and 254 nm; (3) HA-2 achieved a higher removal of typical cyclic compounds, including refractory aromatic substances. Furthermore, a dramatic reduction in the acute toxicity and improvement of the biodegradability was obtained in HA-2. Meanwhile, the "indirect estrogenic activity" in the CGW should cause concern for researchers. A Pearson correlation analysis revealed that spectral indicators could be alternatives for revealing the variation in the regular parameters (i.e. COD and TPh) and the acute toxicity during the anaerobic treatment process of CGW, and a significant correlation was observed between the aromatic compounds and acute toxicity. This study demonstrated that spectral indicators showed high potential in monitoring the performance of practical CGW treatment.

CRediT authorship contribution statement

Zhiqiang Chen: Writing - review & editing. **Da Li:** Writing - original draft. **Qinxue Wen:** Conceptualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Science Foundation of China (51878214) and the National Key Research and Development Program of China (2019YFD1100200).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2020.137995.

References

Abbasi, T., Abbasi, S.A., 2012. Formation and impact of granules in fostering clean energy production and wastewater treatment in upflow anaerobic sludge blanket (UASB) reactors. Renew. Sust. Energ. Rev. 16 (3), 1696–1708.
Aftab, B., Hur, J., 2019. Unraveling complex removal behavior of landfill leachate upon the treatments of Fenton oxidation and MIEX® via two-dimensional correlation size exclusion chromatography (2D-CoSEC). J. Hazard. Mater. 362, 36–44.
Aftab, B., Shin, H., Hur, J., 2018. Exploring the fate and oxidation behaviors of different organic constituents in landfill leachate upon Fenton oxidation processes using EEM-PARAFAC and 2D-COS-FTIR. J. Hazard. Mater. 354, 33–41.
Carstea, E.M., Bridgeman, J., Baker, A., Reynolds, D.M., 2016. Fluorescence spectroscopy for wastewater monitoring: a review. Water Res. 95, 205–219.
Chen, W., Westerhoff, P., Leenheer, J.A., Booksh, K., 2003. Fluorescence excitation—emission matrix regional integration to quantify spectra for dissolved organic matter. Environ. Sci. Technol. 37, 5701–5710.
Chen, Y., Cheng, J.J., Creamer, K.S., 2008. Inhibition of anaerobic digestion process: a review. Bioresour. Technol. 99, 4044–4064.
Chen, Z., Li, M., Wen, Q., 2017a. Comprehensive evaluation of three sets of advanced wastewater treatment trains for treating secondary effluent: organic micropollutants and bio-toxicity. Chemosphere 189, 426–434.
Chen, Z., Li, M., Wen, Q., Ren, N., 2017b. Evolution of molecular weight and fluorescence of effluent organic matter (EfOM) during oxidation processes revealed by advanced spectrographic and chromatographic tools. Water Res. 124, 566–575.
Chen, Q., Wu, W., Qi, D., Ding, Y., Zhao, Z., 2020. Review on microaeration-based anaerobic digestion: state of the art, challenges, and prospectives. Sci. Total Environ. 710, 136388.
Chinese, S.E.P.A.C, 2002. Standard Methods for the Examination of Water and Wastewaters. fourth ed. China Environmental Science Press, Beijing.
Du, Y., Wu, Q., Lv, X., Ye, B., Zhan, X., Lu, Y., Hu, H., 2018. Electron donating capacity reduction of dissolved organic matter by solar irradiation reduces the cytotoxicity formation potential during wastewater chlorination. Water Res. 145, 94–102.
Fuchs, G., Boll, M., Heider, J., 2011. Microbial degradation of aromatic compounds — from one strategy to four. Nat. Rev. Microbiol. 9, 803.
Guo, X., He, X., Li, C., Li, N., 2019. The binding properties of copper and lead onto compost-derived DOM using Fourier-transform infrared, UV–vis and fluorescence spectra combined with two-dimensional correlation analysis. J. Hazard. Mater. 365, 457–466.
Hur, J., Jung, K., Jung, Y.M., 2011. Characterization of spectral responses of humic substances upon UV irradiation using two-dimensional correlation spectroscopy. Water Res. 45, 2965–2974.
Ji, Q., Tabassum, S., Hena, S., Silva, C.G., Yu, G., Zhang, Z., 2016. A review on the coal gasification wastewater treatment technologies: past, present and future outlook. J.Clean. Prod. 126, 38–55.
Kong, Z., Li, L., Li, Y., 2019a. Long-term performance of UASB in treating N, N-dimethylformamide-containing wastewater with a rapid start-up by inoculating mixed sludge. Sci. Total Environ. 648, 1141–1150.
Kong, Z., Li, L., Xue, Y., Yang, M., Li, Y., 2019b. Challenges and prospects for the anaerobic treatment of chemical-industrial organic wastewater: a review. J. Clean. Prod. 231, 913–927.
Korak, J.A., Dotson, A.D., Summers, R.S., Rosario-Ortiz, F.L., 2014. Critical analysis of commonly used fluorescence metrics to characterize dissolved organic matter. Water Res. 49, 327–338.
Li, M., Chen, Z., Wang, Z., Wen, Q., 2019. Investigation on degradation behavior of dissolved effluent organic matter, organic micro-pollutants and bio-toxicity reduction from secondary effluent treated by ozonation. Chemosphere 217, 223–231.

Liu, X., Chen, Q., Zhu, L., 2016. Improving biodegradation potential of domestic wastewater by manipulating the size distribution of organic matter. J. Environ. Sci. 47, 174–182.
Ma, W., Han, Y., Xu, C., Han, H., Ma, W., Zhu, H., Li, K., Wang, D., 2018a. Enhanced degradation of phenolic compounds in coal gasification wastewater by a novel integration of micro-electrolysis with biological reactor (MEBR) under the micro-oxygen condition. Bioresour. Technol. 251, 303–310.
Ma, W., Han, Y., Xu, C., Han, H., Zhu, H., Li, K., Zheng, M., 2018b. Biotoxicity assessment and toxicity mechanism on coal gasification wastewater (CGW): a comparative analysis of effluent from different treatment processes. Sci. Total Environ. 637–638, 1–8.
Ma, X.Y., Wang, Y., Dong, K., Wang, X.C., Zheng, K., Hao, L., Ngo, H.H., 2019a. The treatability of trace organic pollutants in WWTP effluent and associated biotoxicity reduction by advanced treatment processes for effluent quality improvement. Water Res. 159, 423–433.
Ma, S., Ma, H., Hu, H., Ren, H., 2019b. Effect of mixing intensity on hydrolysis and acidification of sewage sludge in two-stage anaerobic digestion: characteristics of dissolved organic matter and the key microorganisms. Water Res. 148, 359–367.
Meckenstock, R.U., Mouttaki, H., 2011. Anaerobic degradation of non-substituted aromatic hydrocarbons. Curr. Opin. Biotechnol. 22, 406–414.
Na, C., Zhang, Y., Quan, X., Chen, S., Liu, W., Zhang, Y., 2017. Evaluation of the detoxification efficiencies of coking wastewater treated by combined anaerobic-anoxic-oxic (A^2O) and advanced oxidation process. J. Hazard. Mater. 338, 186–193.
Nishikawa, J.I., Saito, K., Goto, J., Dakeyama, F., Matsuo, M., Nishihara, T., 1999. New screening methods for chemicals with hormonal activities using interaction of nuclear hormone receptor with coactivator. Toxicol. Appl. Pharmacol. 154, 76–83.
Noda, I., 2012. Close-up view on the inner workings of two-dimensional correlation spectroscopy. Vib. Spectrosc. 60, 146–153.
Novak, M., Žegura, B., Modic, B., Heath, E., Filipič, M., 2017. Cytotoxicity and genotoxicity of anticancer drug residues and their mixtures in experimental model with zebrafish liver cells. Sci. Total Environ. 601-602, 293–300.
Peng, S., He, X., Pan, H., 2018. Spectroscopic study on transformations of dissolved organic matter in coal-to-liquids wastewater under integrated chemical oxidation and biological treatment process. J. Environ. Sci. 70, 206–216.
Persoone, G., Marsalek, B., Blinova, I., Törökne, A., Zarina, D., Manusadzianas, L., Nalecz-Jawecki, G., Tofan, L., Stepanova, N., Tothova, L., Kolar, B., 2003. A practical and user-friendly toxicity classification system with microbiotests for natural waters and wastewaters. Environ. Toxicol. 18, 395–402.
Phong, D.D., Hur, J., 2016. Non-catalytic and catalytic degradation of effluent dissolved organic matter under UVA-and UVC-irradiation tracked by advanced spectroscopic tools. Water Res. 105, 199–208.
Ramakrishnan, A., Surampalli, R.Y., 2012. Comparative performance of UASB and anaerobic hybrid reactors for the treatment of complex phenolic wastewater. Bioresour. Technol. 123, 352–359.
Shi, J., Han, Y., Xu, C., Han, H., 2019a. Anaerobic bioaugmentation hydrolysis of selected nitrogen heterocyclic compound in coal gasification wastewater. Bioresour. Technol. 278, 223–230.
Shi, J., Xu, C., Han, Y., Han, H., 2019b. Enhanced anaerobic biodegradation efficiency and mechanism of quinoline, pyridine, and indole in coal gasification wastewater. Chem. Eng. J. 361, 1019–1029.
Show, K., Yan, Y., Yao, H., Guo, H., Li, T., Show, D., Chang, J., Lee, D., 2020. Anaerobic granulation: a review of granulation hypotheses, bioreactor designs and emerging green applications. Bioresour. Technol. 300, 122751.
Song, G., Yu, Y., Liu, T., Xi, H., Zhou, Y., 2019. Performance of microaeration hydrolytic acidification process in the pretreatment of 2-butenal manufacture wastewater. J. Hazard. Mater. 369, 465–473.
Stedmon, C.A., Bro, R., 2008. Characterizing dissolved organic matter fluorescence with parallel factor analysis: a tutorial. Limnol. Oceanogr. Methods 6, 572–579.
Tang, X., Wu, Q., Zhao, X., Du, Y., Huang, H., Shi, X., Hu, H., 2014. Transformation of anti-estrogenic-activity related dissolved organic matter in secondary effluents during zonation. Water Res. 48, 605–612.
Vojta, D., Vazdar, M., 2014. The study of hydrogen bonding and π···π interactions in phenol···ethynylbenzene complex by IR spectroscopy. Spectrochim. Acta A 132, 6–14.
Wang, L., Hu, H., Wang, C., 2007. Effect of ammonia nitrogen and dissolved organic matter fractions on the genotoxicity of wastewater effluent during chlorine disinfection. Environ. Sci. Technol. 41, 160–165.
Wang, W., Ma, W., Han, H., Li, H., Yuan, M., 2011. Thermophilic anaerobic digestion of Lurgi coal gasification wastewater in a UASB reactor. Bioresour. Technol. 102, 2441–2447.
Wang, K., Li, W., Gong, X., Li, X., Liu, W., He, C., Wang, Z., Minh, Q.N., Chen, C., Wang, J., 2014. Biological pretreatment of tannery wastewater using a full-scale hydrolysis acidification system. Int. Biodeterior. Biodegradation 95, 41–45.
Wen, Q., Ji, Y., Hao, Y., Huang, L., Chen, Z., Sposob, M., 2018. Effect of sodium chloride on polyhydroxyalkanoate production from food waste fermentation leachate under different organic loading rate. Bioresour. Technol. 267, 133–140.
Wu, Q., Hu, H., Zhao, X., Sun, Y., 2009. Effect of chlorination on the estrogenic/Antiestrogenic activities of biologically treated wastewater. Environ. Sci. Technol. 43, 4940–4945.
Wu, C., Zhou, Y., Wang, P., Guo, S., 2015. Improving hydrolysis acidification by limited aeration in the pretreatment of petrochemical wastewater. Bioresour. Technol. 194, 256–262.
Wu, C., Zhou, Y., Sun, Q., Fu, L., Xi, H., Yu, Y., Yu, R., 2016. Appling hydrolysis acidification-anoxic–oxic process in the treatment of petrochemical wastewater: from bench scale reactor to full scale wastewater treatment plant. J. Hazard. Mater. 309, 185–191.
Xu, P., Han, H., Hou, B., Jia, S., Zhao, Q., 2015. Treatment of coal gasification wastewater by a two-phase anaerobic digestion. Desalin. Water Treat. 54, 598–608.
Xu, P., Xu, H., Zheng, D., 2019. The efficiency and mechanism in a novel electro-Fenton process assisted by anodic photocatalysis on advanced treatment of coal gasification wastewater. Chem. Eng. J. 361, 968–974.
Zhang, M., Zhi, L., Li, H., Long, H., Zhao, W., 2013. Process integration of halogenation and oxidation for recovery and removal of phenols from high strength phenolic wastewater. Chem. Eng. J. 229, 99–104.
Zhang, Z., Gao, P., Cheng, J., Liu, G., Zhang, X., Feng, Y., 2018. Enhancing anaerobic digestion and methane production of tetracycline wastewater in EGSB reactor with GAC/NZVI mediator. Water Res. 136, 54–63.
Zhang, B., Shan, C., Hao, Z., Liu, J., Wu, B., Pan, B., 2019. Transformation of dissolved organic matter during full-scale treatment of integrated chemical wastewater: molecular composition correlated with spectral indexes and acute toxicity. Water Res. 157, 472–482.
Zhao, Q., Liu, Y., 2016. State of the art of biological processes for coal gasification wastewater treatment. Biotechnol. Adv. 34, 1064–1072.
Zheng, M., Han, Y., Xu, C., Han, H., Zhang, Z., 2018. Discrimination of typical cyclic compounds and selection of toxicity evaluation bioassays for coal gasification wastewater (CGW) based on toxicity mechanism of actions (MOAs). Sci. Total Environ. 644, 324–334.
Zheng, M., Han, Y., Xu, C., Zhang, Z., Han, H., 2019. Selective adsorption and bioavailability relevance of the cyclic organics in anaerobic pretreated coal pyrolysis wastewater by lignite activated coke. Sci. Total Environ. 653, 64–73.
Zhu, H., Ma, W., Han, H., Han, Y., Ma, W., 2017. Catalytic ozonation of quinoline using nano-MgO: efficacy, pathways, mechanisms and its application to real biologically pretreated coal gasification wastewater. Chem. Eng. J. 327, 91–99.
Zhu, H., Han, Y., Xu, C., Han, H., Ma, W., 2018. Overview of the state of the art of processes and technical bottlenecks for coal gasification wastewater treatment. Sci. Total Environ. 637-638, 1108–1126.